

Mechanical Design Description

Detailed mechanical design of Boaty Mk1 in CadQuery: parts, assembly, mass, hydrostatics and compliance, for the Critical Design Review

Document	BOATY-MDD-001
Issue	Issue A (draft for CDR)
Date	30 September 2026
Baseline	Upstream documents as tagged sdr-baseline-1 (unchanged)
CAD	mechanical/ at commit c856b2a
Compliance	33 pass, 3 by test, 4 not met (CR-09 to CR-11 proposed)

Document control

Issue	Date	Change	By
A	30 September 2026	First issue, draft for the CDR.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review this document with the CAD zip open. Section 2 has the decisions; section 9 has every check; section 10 has the change requests.

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1. Introduction

1.1 Purpose and status

This document describes the detailed mechanical design of Boaty Mk1: every part, how the parts go together, and the analysis that shows whether the design meets the mechanical requirements allocated to it. It is the mechanical input to the Critical Design Review (CDR), which the owner has chosen to hold before anything is bought.

It is a **draft for the CDR**. The system documentation was baselined at the SDR (tag sdr-baseline-1) and is not changed here. Where the design cannot meet a baselined requirement, section 10 proposes a numbered change request for the owner to accept or reject at the CDR.

1.2 Scope

In scope: the hull and structure (SSS-HUL), the thruster pods' mechanical parts (SSS-PRP), the recovery and signalling hardware (SSS-REC), and the mechanical interfaces IF-16 to IF-20. Not in scope: the electronics inside the box, wiring and connector selection (these are modelled only as masses and envelopes), and the bank kit.

1.3 The CAD

The design is a parametric CadQuery model (Python). One file, mechanical/boaty_cad/params.py, holds every dimension with the requirement it answers. The part builders, the assembly, the mass properties, the hydrostatics and the compliance checks all read it. mechanical/build_cad.py rebuilds everything: it writes STEP and STL files, the figures in this document, and the results file this document is built from. This issue was built from commit c856b2a with CadQuery 2.8.0.

Frame: **x** forward from the stern transom, **y** to port from the boat centreline, **z** up from the keel line (the flat bottom of the mid and stern segments). All dimensions are in mm, masses in g.

1.4 References

Ref	Document
[1]	BOATY-SRS-001 Issue H System Requirements Specification
[2]	BOATY-ADD-001 Issue H Architecture Design Document
[3]	BOATY-ICD-001 Issue H Interface Control Document (IF-16 to IF-20)
[4]	Subsystem specifications: HUL B, PRP C, PWR D, HLM F, MCP F, MCN F, REC B, SIM G
[5]	BOATY-KCL-001 Issue D Key Component List
[6]	BOATY-SDR-001 Issue D System Design Review
[7]	EN 71-1 small-parts cylinder (Ø31.7 mm, 25.4 to 57.1 mm deep)

2. Summary

The design meets **33** of the 40 checks in the compliance matrix (section 9). **3** can only be shown by test on the first prints and are listed with their verification. **4** are not met; each has a proposed change request in section 10.

Quantity	Design	Requirement
Mass, ready to sail	2165 g	≤ 1800 g (MEC-002)
Design mass (+ 300 g on the deck)	2465 g	sized at 2100 g
Length $\times$ beam $\times$ height	615 $\times$ 354 $\times$ 492 mm	$\leq 620 \times 360 \times 500$ (HUL-D01)
Mast off	615 $\times$ 354 $\times$ 247 mm	$\leq 650 \times 400 \times 250$ (HUL-D21)
Hull draft; least freeboard (design mass)	34.4; 50.7 mm	≤ 35 ; ≥ 50 (HUL-D07)
Trim light / design (bow down –)	-0.8° / -1.7°	$\pm 3^\circ$ (HUL-D20)
Heel, 300 g at the deck edge	2.1°	$\leq 10^\circ$ (HUL-D08)

Quantity	Design	Requirement
Max righting arm; vanishing angle	110 mm at 15°; 65°	no capsize (HUL-D09)
Reserve buoyancy (2 segments breached)	7587 g	≥ 3150 g (HUL-D06)
Largest guard opening	5.75 mm	< 8 mm probe (MEC-010)
GNSS to power wiring	252 mm	≥ 150 mm (MEC-013)
Fluorescent share of the top view	64%	≥ 50% (REC-D01)

Decisions for the owner at the CDR.

- M-01 MEC-002: Ready-to-sail mass, excluding payload is 2165 g against ≤ 1800 g → **CR-09**
- M-02 HUL-D03: Hull and structure mass is 1551 g against ≤ 1000 g → **CR-09**
- G-08 HUL-D10, MEC-006: Hull shell: material, perimeters, wall is ASA; 3 perimeters; 1.2 mm against PETG/ASA; ≥ 3 ; ≥ 1.6 mm → **CR-11**
- E-02 IF-19: Masthead top above the water (design mass) is 415 mm against ≈ 450 , total height ≤ 500 → **CR-10**

The one that matters is mass (CR-09): the boat comes out about 20% heavier than the SRS allows, almost all of it in the hull. Everything that depends on mass (draft, freeboard, heel, stability, trim) has been checked at the real mass and passes.

3. General arrangement

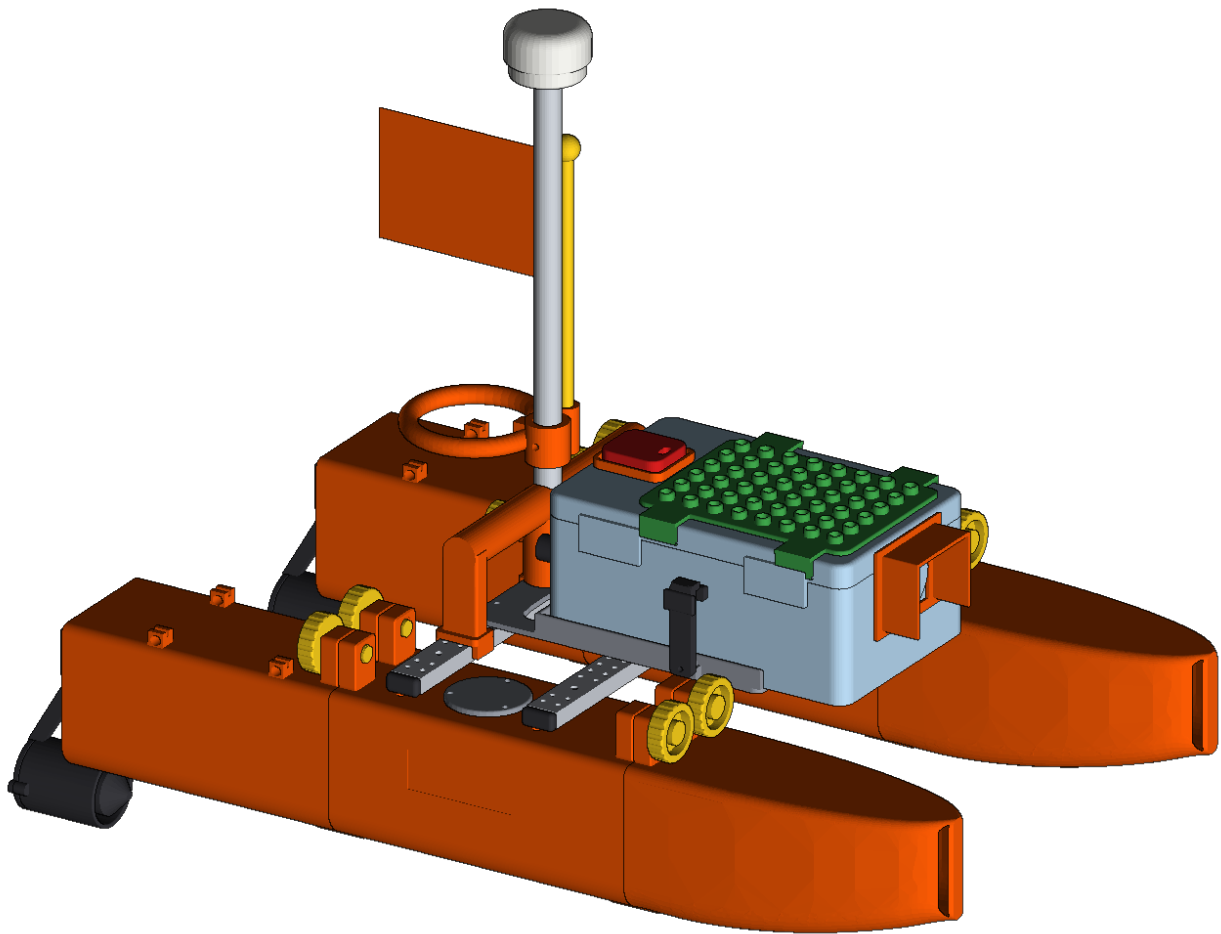


Figure 1. Boaty Mk1, from forward to starboard. Fluorescent orange: hull segments, handle, hood, hoop (REC-D01). Green: DUPLO deck. Clear: the bought box.

Two hulls of three foam-cored segments each sit 270 mm apart. Two aluminium crossbeams bolt to the mid segments only, so the mid segments, the beams and everything on them form a rigid core. The bow and stern segments and the pods clip onto that core without tools: these are the crew's assembly steps (HUL-D25). The box rides on a saddle tray across both beams. The mast stands in a socket on the aft beam, and the socket part is also the carry handle, 39 mm from the centre of gravity.

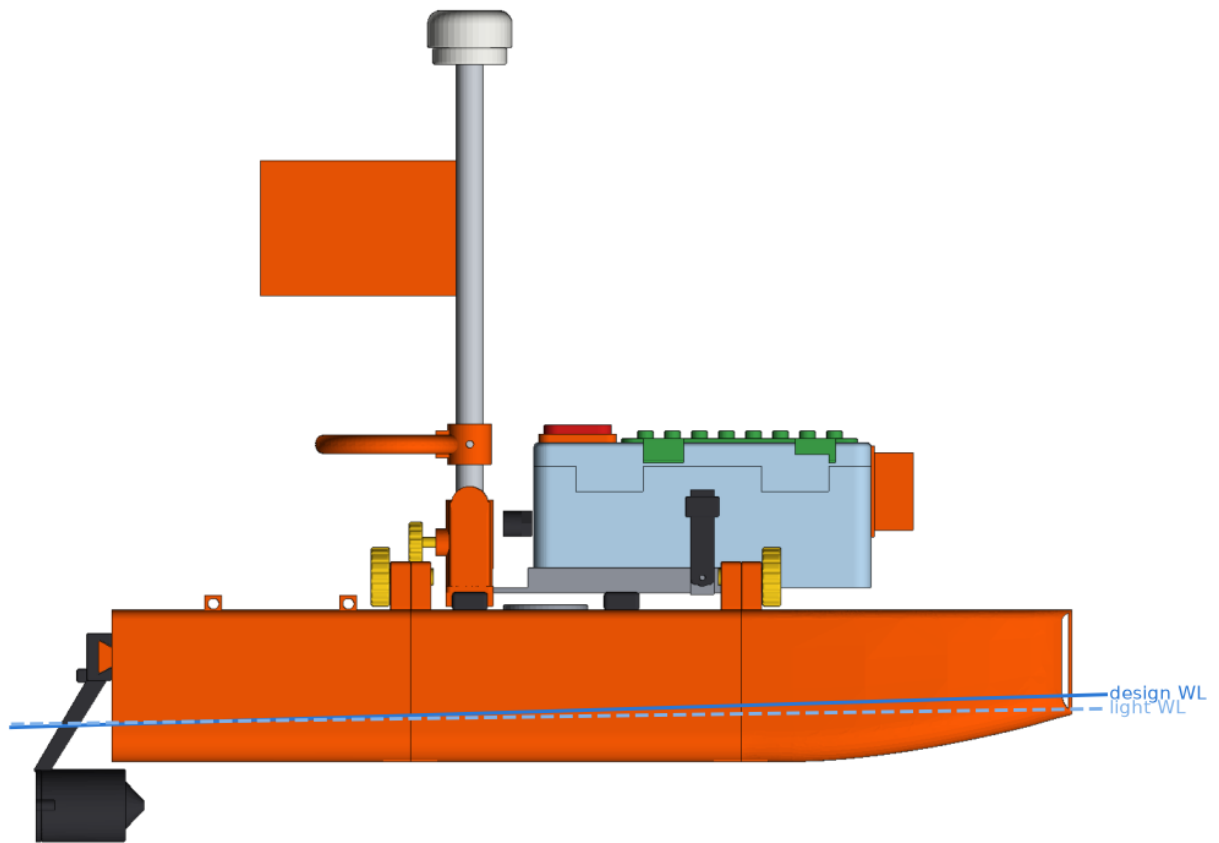


Figure 2. Side elevation from starboard, with the design waterline (2465 g, trim -1.7°) and the light waterline (2165 g).

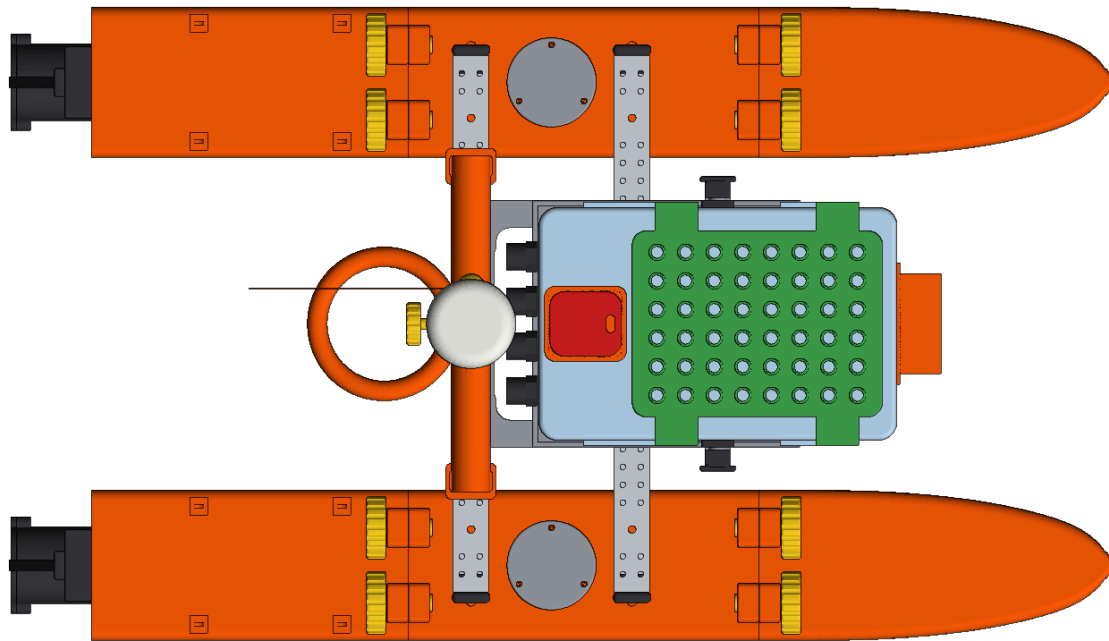


Figure 3. Plan view (the view the REC-D01 fluorescent share is measured on).

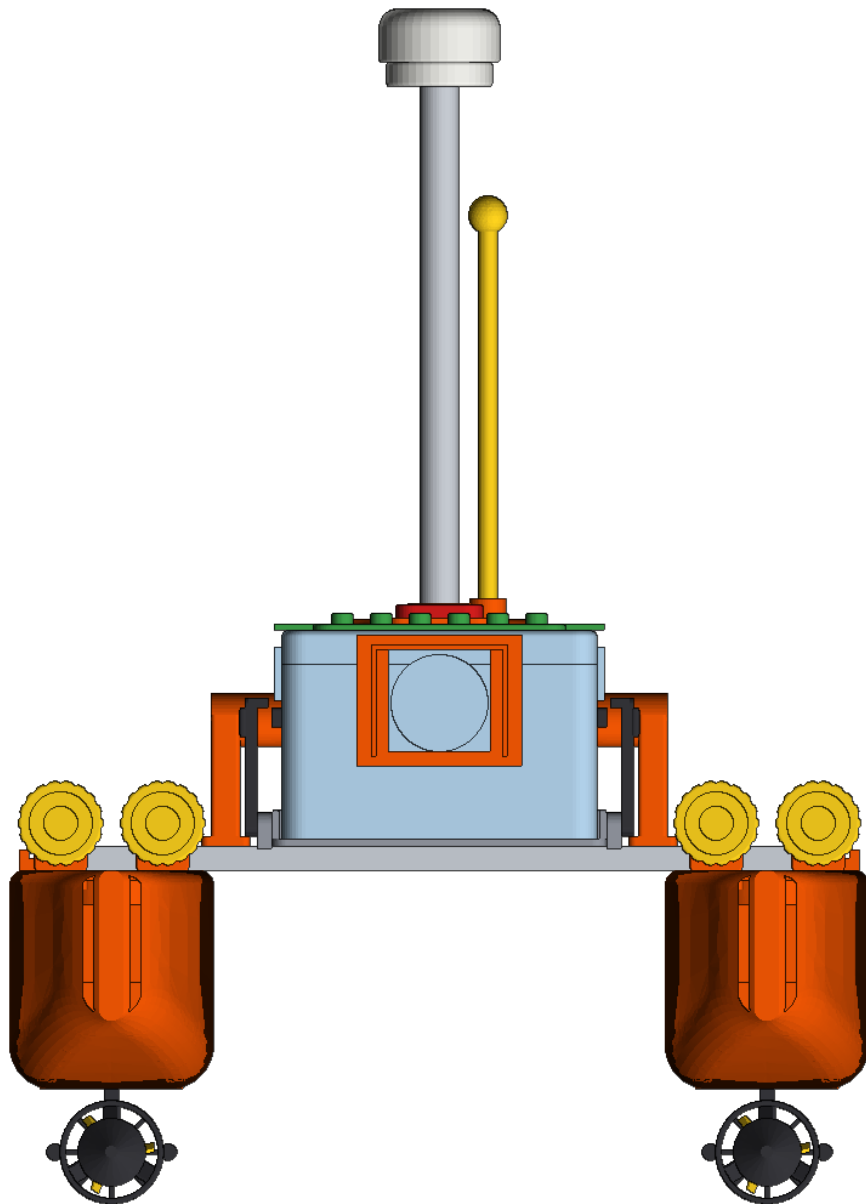


Figure 4. From ahead: the camera hood, the pods under the sterns, the masthead.

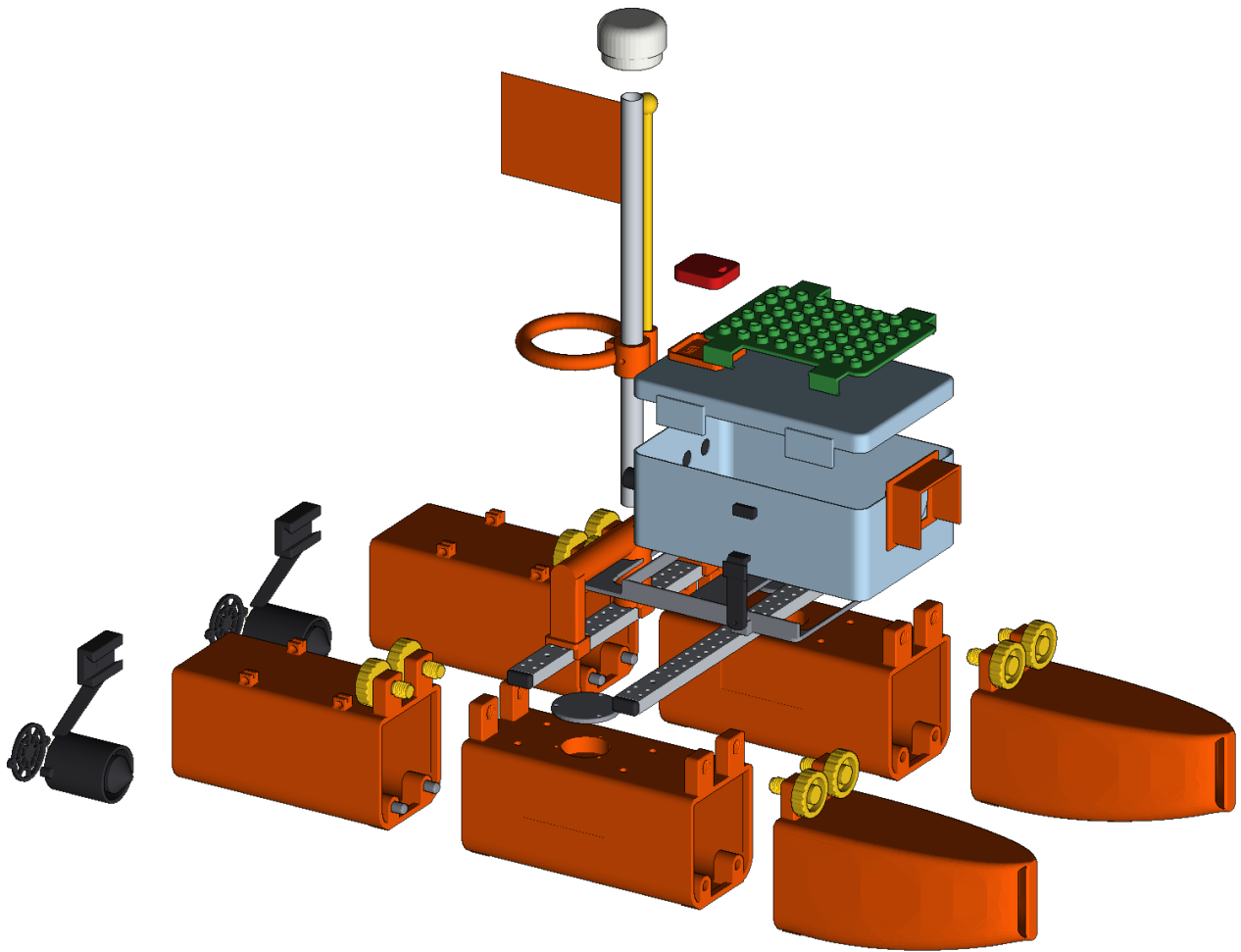


Figure 5. Exploded: end segments, pods, beams, tray, box, lid, deck and mast.

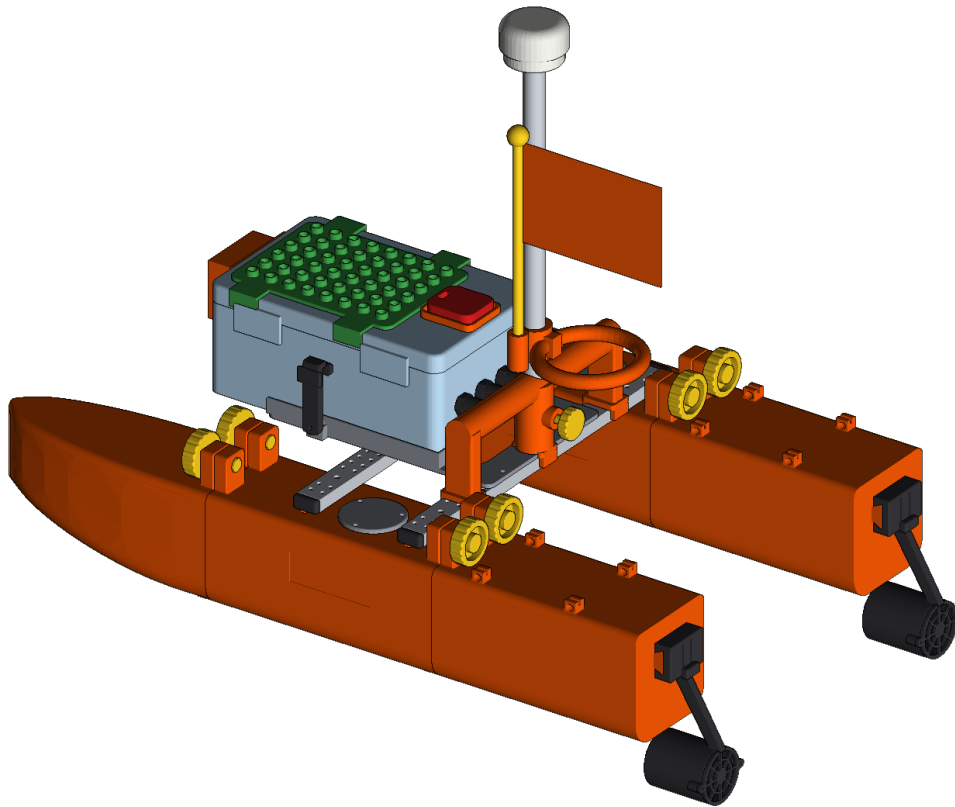


Figure 6. From aft to port: pod guards, the glands on the box's aft face, the hoop.

3.1 Main dimensions

Item	Value	Source
Hull section at the joints (IF-16)	84 wide × 90 deep; bilge radius 18	closes TBC-11
Segment lengths stern / mid / bow	177 / 196 / 196	HUL-D02
Bow keel rise at the stem	28	weed and wave entry
Hull centreline spacing	270	HUL-D04
Crossbeams (x)	212, 302	on the mid segments
Box, external	200 × 130 × 86, aft face at x 250	IF-18
Mast	Ø16 × 1 Al at x 212	IF-19
Prop axis; duct	z -26; Ø38 / 42.8, x -42 to 8	IF-17
Centre of gravity (light)	251, 0, 98	section 5

4. Part design

4.1 Hull segments (HUL-1, IF-16)

Each segment is a thin ASA shell, open at every joint face, with a core of closed-cell XPS foam profiled to the section (hot-wire cut from 50 mm board, bonded in two layers, gaps filled with PU foam) and trimmed flush with the joint face. The foam is the core of a sandwich: it stabilises the 1.2 mm skin and gives the buoyancy that survives a breach (HUL-D05/D06). Each of the three designs is symmetric about its own centreline, so the same part serves both hulls.

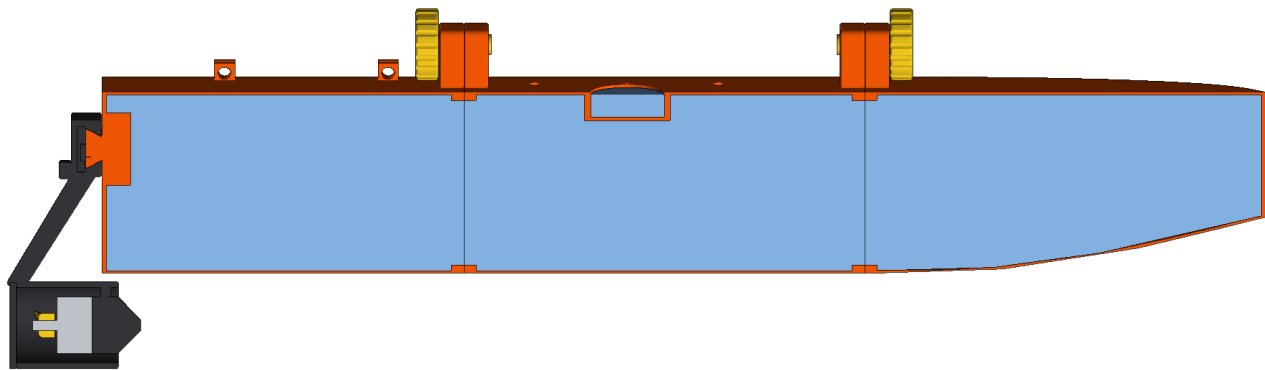


Figure 7. One hull cut on its centreline: shells (orange), foam cores (blue), the thumb-screw lugs at both joints (the dowels lie off this plane), the dovetail rail and the pod.

Lines. Mid and stern: a constant section, 84 × 90 mm with an 18 mm bilge and a 6 mm deck edge. The bow keeps the full section for 30 mm, then narrows to a 10 mm rounded stem while the keel rises 28 mm, so weed and small waves ride under it.

Joint (IF-16). The last 6 mm of each open end is a 4 mm flange; the mating faces are flat. Two Ø8 dowels, bonded 10 mm into the bow and stern segments, locate in 8.3 mm holes in the mid segment low in the section. Two captive M12 × 1.75 printed thumb-screws with Ø35 knurled knobs sit on the deck, 7 mm apart: the knob is on the end segment, whose lug is threaded so the screw threads through it once and then turns freely on its Ø9 neck (it cannot fall out, CHD-001), and screws into the mid segment's lug. Hand-tight (20 N at the rim) they clamp the joint at about 97 N each, so a joint carries 15.5 N·m of sag before it opens, against 3.4 N·m at three times design weight with the boat supported at its ends.

Other features. Mid segment: four M4 heat-set insert bosses under the beams; the REC-D05 tracker pocket (Ø36 × 12, O-ring cap on three M3 screws); a 70 × 24 label recess on each side (REC-D04). Stern segment: the IF-17 dovetail rail on a solid backing block in the transom, and two cable clips on each deck edge for the motor lead.

Part	Name	Qty	Kind	Material	Size (mm)
HUL-101	Bow segment	2	printed	ASA	204 × 88 × 120
HUL-102	Mid segment	2	printed	ASA	196 × 84 × 119
HUL-103	Stern segment	2	printed	ASA	185 × 84 × 119
HUL-104	Joint thumb-screw M12	8	printed	PETG	35 × 35 × 37
HUL-105	Joint dowel Ø8 × 20 (bonded)	8	printed	PETG	8 × 8 × 20
HUL-106	Tracker pocket cap	2	printed	PETG	50 × 50 × 3

4.2 Crossbeams (HUL-2)

20 × 10 × 1.5 mm 6063 aluminium rectangular tube, 300 long, lying flat on the mid segments' decks. Two M4 A4 screws at each end go into heat-set inserts in the segment (an adult, tool step: the beams are not part of the crew's assembly). The top face carries the M3 rail grid (two rows at 10 mm pitch) named in SSS-HUL; tray and handle bolt through it with nylon-insert nuts under the tube. Printed caps are bonded into the tube ends. A printed beam would not fit the 200 mm build volume in one piece.

Part	Name	Qty	Kind	Material	Size (mm)
HUL-201	Crossbeam 20 × 10 × 1.5 Al tube, 300 long	2	stock	Al 6063	20 × 300 × 10
HUL-202	Beam end cap (bonded)	4	printed	PETG	21 × 16 × 11

4.3 Electronics box, saddle and latches (HUL-3, IF-18)

The box is a bought clip-lock food box, modelled at 200 × 130 × 86 external, 1.6 mm walls. Its only modifications are four 12.7 mm holes in the aft face for the PG7 glands at 25 mm pitch, and a Ø30 window hole in the front, covered by a 2 mm acrylic disc bonded outside. The final box choice sets BOX_* in the parameters; the tray follows (TBC-13).

The printed saddle tray (194 long) spans both beams, bolted with four M3 screws through the rail. Side walls and an aft stop locate the box; the box overhangs the tray by 54 mm at the front, over the forward beam. Behind the box a 38 mm gap leaves room for the glands, and a slot in the tray lets the leads drop to the beam. Two over-centre latch levers pivot on M3 pins in bosses on the tray sides and hook keepers bonded to the box. The lever passes centre by 2 mm and has a flexure detent sized for ≥ 40 N at the tab (HUL-D24): an adult action, measured on the bench.

Key dock. A printed cup bonded to the lid over the reed switch, with KEY raised in its floor. The key is a 40 × 36 × 9 mm fob with a Ø20 magnet bonded in and a lanyard slot; at 36 mm across it cannot enter the small-parts cylinder.

Part	Name	Qty	Kind	Material	Size (mm)
HUL-301	Electronics box body (clip-lock, modified)	1	bought	PP	200 × 130 × 72
HUL-302	Electronics box lid	1	bought	PP	200 × 136 × 29
HUL-303	Saddle tray	1	printed	PETG	194 × 150 × 16
HUL-304	Over-centre latch	2	printed	PETG	20 × 12 × 57
HUL-305	Latch keeper (bonded to box)	2	printed	PETG	18 × 4 × 8
HUL-306	PG7 cable gland	4	bought	PA	17 × 17 × 15
HUL-307	Key dock (bonded to lid)	1	printed	PETG	46 × 42 × 5
HUL-308	Arming key fob (magnet bonded in)	1	printed	PETG	40 × 36 × 9

4.4 Camera hood (HUL-5)

A three-sided hood bonded to the box front around the window, projecting 22 mm (HUL-D19 asks ≥ 15). It is open at the front and underneath, so the window can be wiped with a cloth and rain drains out. The lens centre is 56 mm above the box floor: 123 mm above the water at design mass. The camera faces along the keel line, so its tilt is the boat's trim.

Part	Name	Qty	Kind	Material	Size (mm)
HUL-501	Camera hood	1	printed	PETG	25 × 68 × 54
HUL-502	Camera window, 2 mm acrylic	1	stock	acrylic	2 × 40 × 40
KC-11	Camera OV5647 (reference)	1	ref	-	12 × 25 × 24

4.5 DUPLO deck (IF-20)

A 2.4 mm plate with 6 × 8 studs at 16 mm pitch, Ø9.4 × 4.5 (TBC-14: tune by test print against genuine bricks). The studs are bored through so rain drains to the lid and runs off (HUL-D13). Four clip arms snap over the lid's long edges: the child lifts the plate off for the adult to open the box, and pushes it back (≤ 20 N, HUL-D25). The plate sits forward on the lid, leaving the key dock clear.

Part	Name	Qty	Kind	Material	Size (mm)
HUL-401	DUPLO deck plate	1	printed	PETG	140 × 136 × 19

4.6 Mast, handle and masthead (HUL-4, IF-19)

One printed part is both the mast step and the carry handle (196 mm across, within the build volume). It sits on the aft beam on three feet, located by cheeks either side of the tube and bolted with M3 screws through the beam. The centre tower is the 50 mm deep IF-19 socket; a captive M8 printed thumb-screw with a Ø24 knob holds the mast. A Ø22 grip runs either side of the tower at 52 mm above the beam, with 41 mm finger clearance. The grip centres are 39 mm; 48 mm from the centre of gravity (HUL-D21: ± 50).

The mast is Ø16 × 1 aluminium tube, 321 long, and is also the conduit for the GNSS and LED lead, which leaves through a slot at the foot of the socket. The masthead is one translucent PETG print: a sleeve on the tube, a ring diffuser over the WS2812 LEDs (REC-D03) and a hollow dome over the GNSS/compass board. The dome's top is 415 mm above the design waterline. IF-19 gives ≈ 450 , but the ≤ 500 mm height limit, measured from the pod bottom, does not leave room for it (CR-10).

Part	Name	Qty	Kind	Material	Size (mm)
HUL-402	Mast step and carry handle	1	printed	PETG	34 × 196 × 71
HUL-403	Mast tube Ø16 × 1 Al, 311 long	1	stock	Al 6063	16 × 16 × 321
HUL-404	Mast thumb-screw M8 (captive)	1	printed	PETG	25 × 24 × 24
REC-104	Masthead: GNSS dome and LED diffuser	1	printed	PETG	50 × 50 × 32

4.7 Recovery and signalling (REC)

Hoop. A Ø64 internal ring of 11 mm round section, horizontal and behind the mast, so a hook on the bank pole drops into it from above. It is part of a collar at z 188, pinned to the tube by one M4 A4 cross-bolt, so it cannot slip or turn. The hoop is low on the mast: a 50 N pull puts 4.3 N·m on the socket, not the 15 N·m the ICD estimated (design 30).

Flag. A Ø8 printed staff, 190 long, with a Ø16 ball cap (REC-D08), plugs into a socket on the collar: this is one of the child's assembly steps. The 120 × 80 fluorescent ripstop flag flies from its top, 326 mm above the water, below the GNSS.

Hi-vis. The hull segments are printed in fluorescent orange ASA; the handle, hood, key dock and hoop in fluorescent PETG. Measured on the rendered plan view, 64% of the visible top is fluorescent (REC-D01 $\geq 50\%$).

Part	Name	Qty	Kind	Material	Size (mm)
REC-101	Collar and recovery hoop	1	printed	PETG	104 × 86 × 28
REC-102	Flag staff with ball cap	1	printed	PETG	16 × 16 × 190
REC-103	Flag, fluorescent ripstop 120 × 80	1	bought	ripstop	120 × 0 × 80

4.8 Thruster pod (PRP-1, IF-17)

The pod slides sideways onto the transom's 60° dovetail rail (20 × 40 mm, 0.3 mm clearance) and a tooth on a flexure tongue in the pod's back wall drops into a detent at the centre of the rail. To release it, an adult pulls the tab aft and slides the pod off. Because the detent is central, one pod design fits either hull from either side.

A strut, its leading edge swept back 32° (PRP-D10), carries the duct (Ø38 inside, 2.4 mm wall) whose top is 4 mm under the keel. The duct runs 8 mm under the hull; the rest is behind the transom. A Ø28 nacelle with a conical nose carries the 2205-class motor on four M2 screws, facing aft. Three inlet vanes at 45° hold the nacelle in the duct. The prop (Ø35, three skewed blades) sits between the bell and the rear guard. The guard is a hub, one ring and eight bars, held by two M2.5 screws: removing it is a tool step (PRP-D14).

Opening a probe could enter	Largest circle (mm)
inlet annulus (nacelle to duct)	5.00
guard cells, hub to ring	3.21
guard cells, ring to duct	5.75

Every opening is under 6 mm, against the 8 mm probe (MEC-010). The pod weighs 73 g with its motor (PRP-D13 $\leq$ 75).

Part	Name	Qty	Kind	Material	Size (mm)
PRP-101	Thruster pod body	2	printed	PETG	65 × 54 × 123
PRP-102	Rear guard	2	printed	PETG	3 × 54 × 43
PRP-103	Propeller Ø35, 3 blades	2	printed	PETG	10 × 30 × 32
KC-04	Motor 2205 (reference)	2	ref	-	29 × 28 × 28

4.9 Part gallery

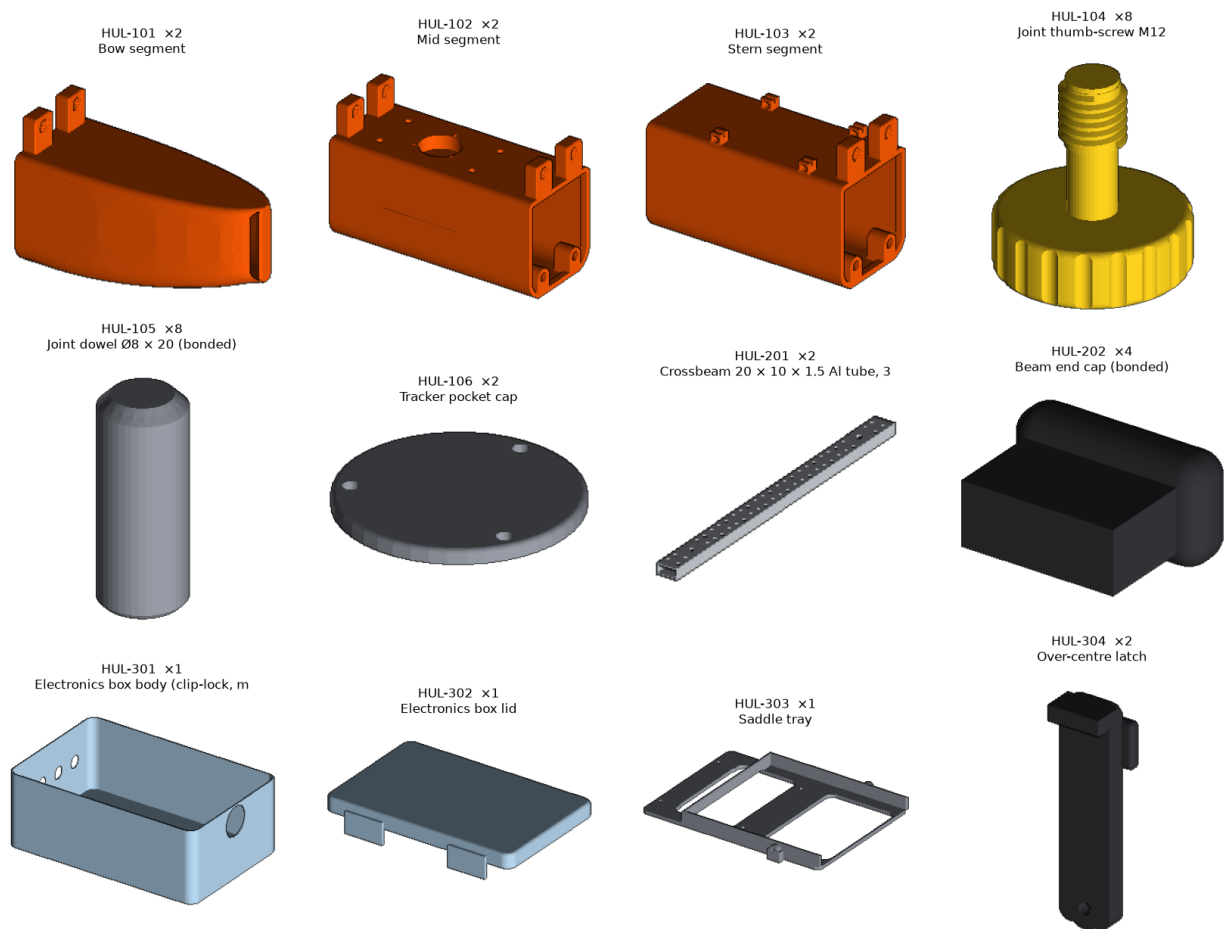


Figure 8. Part designs (1 of 3).

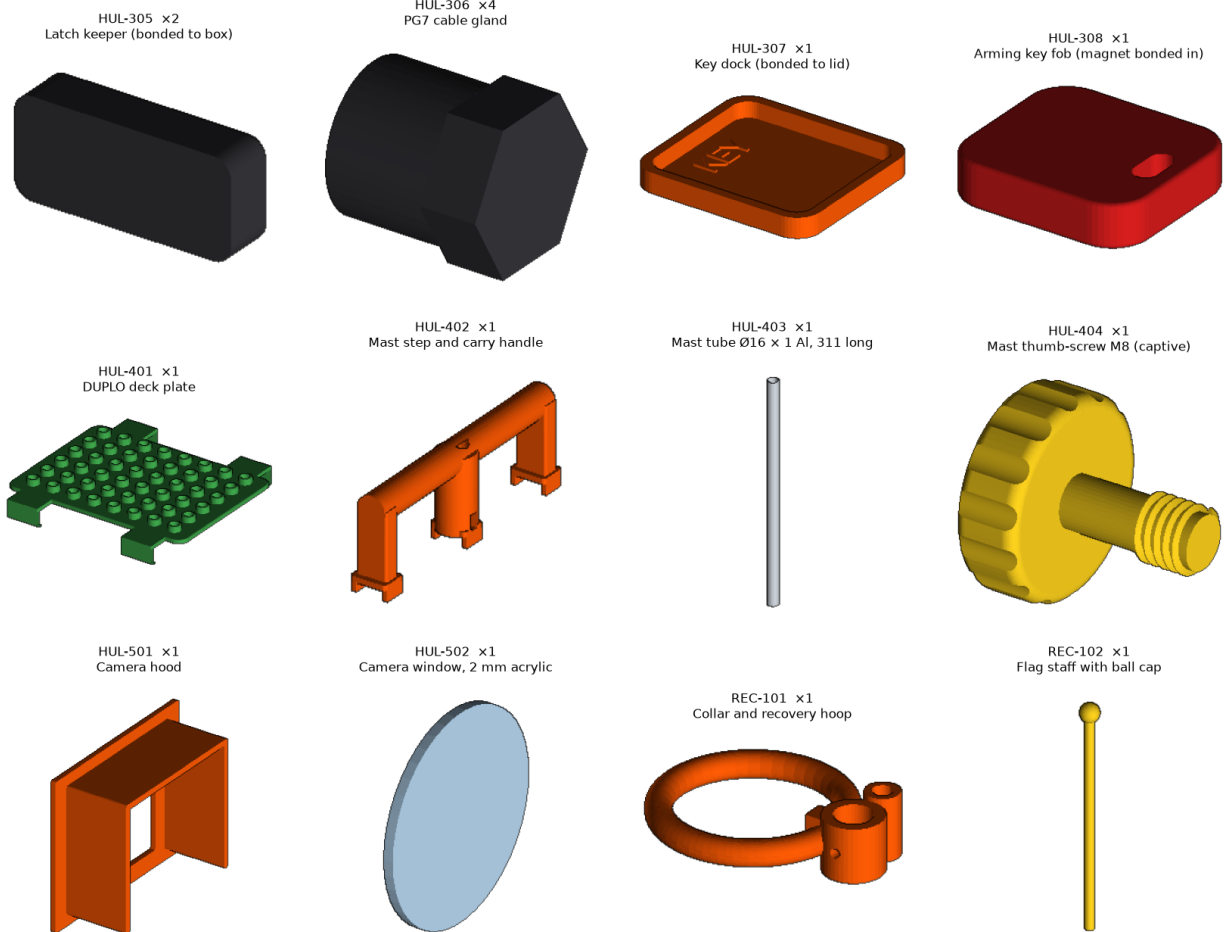
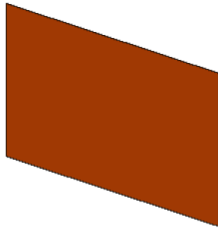
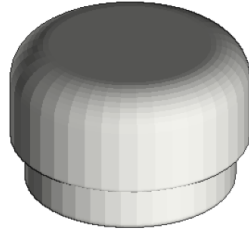


Figure 9. Part designs (2 of 3).

REC-103 x1
Flag, fluorescent ripstop 120 x 80



REC-104 x1
Masthead: GNSS dome and LED diffus



PRP-101 x2
Thruster pod body



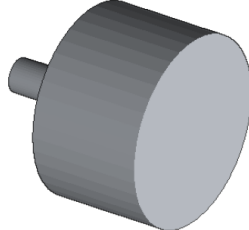
PRP-102 x2
Rear guard



PRP-103 x2
Propeller Ø35, 3 blades



KC-04 x2
Motor 2205 (reference)



KC-11 x1
Camera OV5647 (reference)

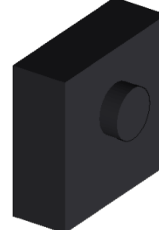


Figure 10. Part designs (3 of 3).

5. Mass properties

Printed parts are weighed as a slicer prints them: walls and top and bottom skins solid, 3 perimeters (1.2 mm), 25% gyroid infill inside solid features (lugs, bosses, blocks). Thin shells come out solid. ASA 1.07, PETG 1.27, XPS foam 0.030, aluminium 2.70 g/cm³. Bought items use catalogue masses; the electronics use the ADD mass table. The electronics are placed in the box with the battery low and aft, which brings the centre of gravity back to the handle and keeps the trim within $\pm 3^\circ$.

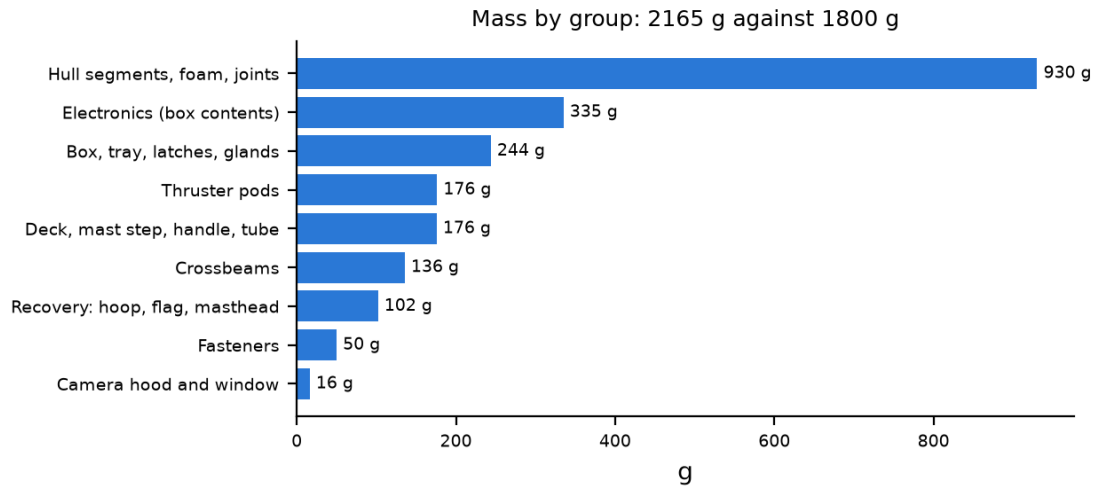


Figure 11. Mass by group.

Item	Qty	Each (g)	Total (g)	Basis
Stern segment	2	156.8	314	CAD, printed 113.2 cm ³ + foam 1.19 L
Mid segment	2	146.0	292	CAD, printed 99.0 cm ³ + foam 1.34 L
Bow segment	2	116.4	233	CAD, printed 79.7 cm ³ + foam 1.04 L
Battery 3S 18650 + BMS	1	160.0	160	ADD A.MASS
Crossbeam 20 × 10 × 1.5 Al tube, 300 long	2	63.7	127	CAD 23.6 cm ³
Electronics box body (clip-lock, modified)	1	95.0	95	catalogue
Thruster pod body	2	41.0	82	CAD, printed 32.3 cm ³
Mast step and carry handle	1	73.3	73	CAD, printed 57.7 cm ³
Joint thumb-screw M12	8	8.7	70	CAD, printed 6.9 cm ³
Saddle tray	1	62.2	62	CAD, printed 49.0 cm ³
Wiring and connectors	1	60.0	60	ADD A.MASS
DUPLO deck plate	1	58.1	58	CAD, printed 45.7 cm ³
Motor 2205 (reference)	2	28.0	56	catalogue
Stainless fasteners, inserts, pins (count in MDD)	1	50.0	50	estimate
Mast tube Ø16 × 1 Al, 311 long	1	40.8	41	CAD 15.1 cm ³
Electronics box lid	1	40.0	40	catalogue
ESCs (2)	1	30.0	30	ADD A.MASS
Power module, buck, key switch	1	30.0	30	ADD A.MASS
Pod connectors (2 pairs) and motor leads	1	30.0	30	estimate
Collar and recovery hoop	1	26.7	27	CAD, printed 21.1 cm ³
Flight controller + PDB	1	25.0	25	ADD A.MASS

Item	Qty	Each (g)	Total (g)	Basis
GNSS + compass board, LED ring, mast cable	1	25.0	25	estimate
Masthead: GNSS dome and LED diffuser	1	22.9	23	CAD, printed 18.0 cm ³
Pi Zero 2 W + SD	1	22.0	22	ADD A.MASS
PG7 cable gland	4	4.0	16	catalogue
Labels, hi-vis tape, O-rings, sealant	1	15.0	15	estimate
Tracker pocket cap	2	7.0	14	CAD, printed 5.5 cm ³
Camera hood	1	13.3	13	CAD, printed 10.4 cm ³
Flag staff with ball cap	1	9.4	9	CAD, printed 7.4 cm ³
Over-centre latch	2	4.6	9	CAD, printed 3.6 cm ³
Arming key fob (magnet bonded in)	1	8.3	8	CAD, printed 6.6 cm ³
Joint dowel Ø8 × 20 (bonded)	8	1.0	8	CAD, printed 0.8 cm ³
Beam end cap (bonded)	4	2.0	8	CAD, printed 1.6 cm ³
Camera OV5647 (reference)	1	8.0	8	catalogue
Magnet (arming key) Ø20 × 3	1	7.0	7	estimate
Rear guard	2	2.8	6	CAD, printed 2.2 cm ³
Key dock (bonded to lid)	1	4.5	4	CAD, printed 3.5 cm ³
Mast thumb-screw M8 (captive)	1	3.5	4	CAD, printed 2.7 cm ³
Camera window, 2 mm acrylic	1	3.0	3	CAD 2.5 cm ³
Flag, fluorescent ripstop 120 × 80	1	3.0	3	catalogue
Propeller Ø35, 3 blades	2	1.4	3	CAD, printed 1.1 cm ³
Latch keeper (bonded to box)	2	0.7	1	CAD, printed 0.6 cm ³
Total			2165	

Centre of gravity (light): x 251, y 0, z 98 mm. Hull and structure (HUL-D03 scope) 1551 g.

Mass is the design's main problem. The boat is 2165 g against 1800 g. The ADD allowed 500 g for the six segments with their foam; the CAD gives 930 g for segments, foam and joint hardware. The shell is set by area, not detail: two hulls of this size have about 0.4 m² of skin, and 1.2 mm of ASA over that is about 500 g before any foam. Section 10 (CR-09) gives the options.

6. Hydrostatics and stability

The underwater body is the two hulls' outer surfaces plus the pods' material (the ducts flood), tessellated at 0.15 mm. For any water plane the displaced volume and its centroid come from signed tetrahedra clipped at the plane, which is exact for the mesh. Sinkage and trim are solved together so that buoyancy equals weight and the trimming moment is zero; heel is then stepped with sinkage and trim free. Fresh water, 1.000 g/cm³. The volume inside the hull envelope is 7.85 L.

Condition	Mass (g)	Trim	Hull draft (mm)	Least freeboard (mm)	Water at transom / stem
Light (ready to sail)	2165	-0.81°	28.8	58.8	23 / 31
Design (+ 300 g on the deck)	2465	-1.72°	34.4	50.7	22 / 39

Draft is the deepest point of the hull below the water; the pods reach deeper (pods reach 78 mm below the water). Water heights are above the keel line; at the stem the keel itself is 28 mm up.

6.1 Heel and stability

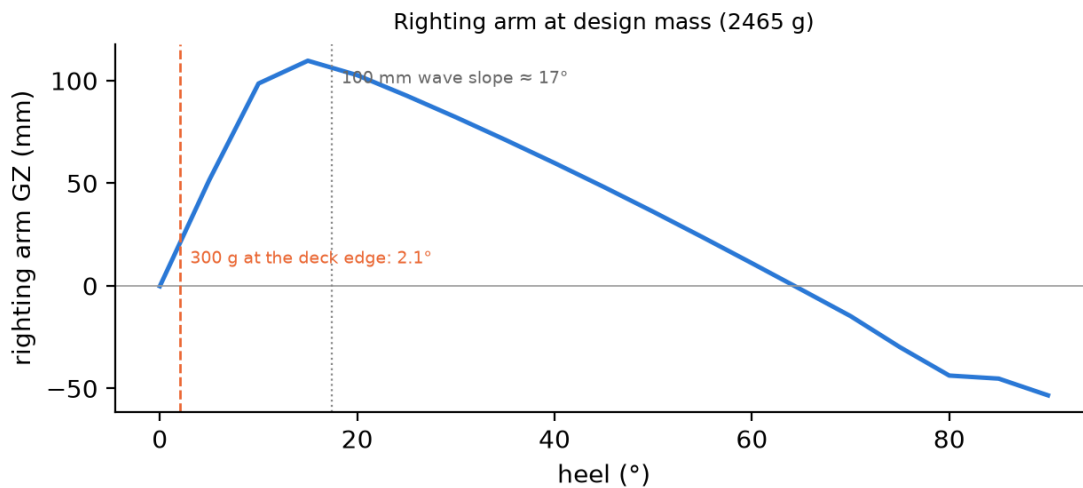


Figure 12. Righting arm at design mass, load on the centreline.

300 g at the outboard deck edge heels the boat 2.1° (MEC-004 ≤ 10°). The righting arm peaks at 110 mm at 15°, when one hull lifts clear, and does not vanish until 65°. The maximum righting moment is 2.65 N·m.

Turns (HUL-D09). A 1.5 m/s turn on a 1 m radius puts 0.54 N·m of heeling moment on the boat (side force at the centre of gravity, resisted at half the draft). That is balanced within 5°; confirmed in the pool (V-item), well short of the peak. **Waves:** a 100 mm wave of 1 m length has a slope of about 17°, far inside the range of positive stability. Both are confirmed in the pool.

Towing (REC-005). Pulled sideways by the hoop, the boat would capsize at 15 N (lever 175 mm). Towed through the water at 0.5 m/s, side-on, the pull equals the drag, about 4.7 N. A 50 N snatch is a strength case for the hoop and socket (section 7), not a steady tow: the operations manual already says to tow slowly.

6.2 Damaged buoyancy

With the box flooded (it gives no buoyancy in any case) and the two largest segments breached, each losing the 10% of its cavity that HUL-D05 allows to be unfilled, the hulls can still support 7587 g: 3.1 times the design mass, against 1.5 (HUL-D06, REC-001).

Heel (°)	0	10	20	30	40	50	60	70	80	90
GZ (mm)	-0	99	103	82	60	36	11	-15	-44	-53

7. Structural checks

Load cases from the ICD: design mass 2.1 kg, factor 3. Here they are taken at the real design mass, 2465 g.

Case	Calculation	Result
Carry by the handle, 3 × design weight (HUL-D21)	Aft beam carries the load at its centre, hulls hang from its ends; 20 × 10 × 1.5 tube	21 MPa against 160 MPa yield
Segment joint, boat supported at bow and stern, 3 × design weight (IF-16)	Hand-tight preload opens the joint only when the sag moment beats preload × lever from the screws to the lower kern	15.5 N·m capacity against 3.4 N·m
50 N on the hoop (REC-005, IF-19)	Moment at the socket foot; hoop ring as a cantilever	4.3 N·m (IF-19 design 30); hoop 29 MPa
Pod: 10 N thrust × 3, 20 N at the tip (IF-17)	Dovetail flanks bear the moment over 20 mm	≈ 100 N on 40 mm of flank: low stress; pull test
Mast tube in the socket	Ø16 × 1 tube at the socket lip under 50 N	≈ 6 MPa

The printed parts' strengths are all well inside PETG's ≈ 50 MPa except the hoop, which the simple cantilever puts close to the limit. The real ring shares the load round both sides, so it is marked for the pull test rather than failed.

8. Child safety and guarding

The child-handled parts (HUL-D22) are the six segments, the thumb-screws, the DUPLO plate, the pod bodies and the flag staff. None fits the EN 71-1 small-parts cylinder in any orientation (CHD-001). The smaller parts that would fit are all bonded, captive, screwed on or inside the box:

Part	Fits the cylinder?	Why it is acceptable
HUL-105 Joint dowel Ø8 × 20 (bonded)	yes	bonded into the segment; never loose
HUL-202 Beam end cap (bonded)	yes	bonded into the beam end
HUL-304 Over-centre latch	yes	pinned to the tray
HUL-305 Latch keeper (bonded to box)	yes	bonded to the box
HUL-306 PG7 cable gland	yes	screwed into the box wall (adult)
PRP-103 Propeller Ø35, 3 blades	yes	inside the guard; tool to reach
KC-11 Camera OV5647 (reference)	yes	inside the sealed box

Edges the child touches carry ≥ 0.6 mm fillets or chamfers in the model: knobs 1.5 mm, deck edges 6 mm, clips and latch tabs 0.6 to 1.5 mm (HUL-D23). Segment joints close face to face, and the knobs sit 7 mm apart with no gap a finger can be caught in when they turn. These are confirmed by inspection on the first prints. The props are behind the inlet annulus and the rear guard, whether a pod is fitted or not (PRP-D14).

9. Compliance matrix

Every check below is computed by boaty_cad/checks.py from the CAD at the commit on the cover.
PASS: met. TEST: the CAD cannot show it; the verification in section 11 will. FAIL: not met; the change request is named.

9.1 Mass

ID	Requirement	Check	Design value	Limit	Status
M-01	MEC-002	Ready-to-sail mass, excluding payload +365 g against the limit	2165 g	≤ 1800 g	FAIL CR-09
M-02	HUL-D03	Hull and structure mass segments, beams, deck, box, mast, fasteners	1551 g	≤ 1000 g	FAIL CR-09
M-03	PRP-D13	Pod mass including the motor body, guard, prop, 28 g motor; lead and connector excluded	73 g	≤ 75 g	PASS

9.2 Geometry and printing

ID	Requirement	Check	Design value	Limit	Status
G-01	HUL-D01, MEC-001	Overall length (hull bow to pod guard)	615 mm	≤ 620 (HUL) / 650 (SRS)	PASS
G-02	HUL-D01, MEC-001	Overall beam	354 mm	≤ 360 / 400	PASS
G-03	HUL-D01, MEC-001	Height, pod bottom to masthead	492 mm	≤ 500	PASS
G-04	HUL-D21, MEC-014	Envelope with the mast removed pods fitted	615 × 354 × 247	$\leq 650 \times 400 \times 250$	PASS
G-05	HUL-D04	Hull centreline spacing	270 mm	≥ 270	PASS
G-06	HUL-D02	Segments per hull and length	3; 177, 196, 196 mm	3; ≤ 200	PASS
G-07	MEC-005, HUL-D02	Largest printed part (fits the 200 mm cube) 22 printed part designs checked	Bow segment: 196 × 119 × 84	every side ≤ 197 (200 less 3 mm margin)	PASS
G-08	HUL-D10, MEC-006	Hull shell: material, perimeters, wall the foam core carries the skin; 1.6 mm adds ≈ 150 g	ASA; 3 perimeters; 1.2 mm	PETG/ASA; ≥ 3 ; ≥ 1.6 mm	FAIL CR-11

9.3 Buoyancy and hydrostatics

ID	Requirement	Check	Design value	Limit	Status
B-01	HUL-D05, MEC-006	Closed-cell foam fill of each segment cavity The CAD fills the cavity; the as-built fill is checked by weighing each filled segment (V-item M-V1)	100% modelled (XPS, profiled blocks)	$\geq 90\%$	TEST
H-01	HUL-D07	Hull draft at design mass (boat + 300 g on the deck) pods reach 78 mm below the water	34.4 mm at 2465 g	≤ 35	PASS
H-02	HUL-D07, MEC-003	Least freeboard at design mass	50.7 mm	≥ 50	PASS
H-03	HUL-D20	Trim (camera level) light / design negative = bow down; lens axis is parallel to the keel	-0.8° / -1.7°	$\pm 3^\circ$	PASS
H-04	HUL-D20	Lens height above the water at design mass	123 mm	≥ 120	PASS

ID	Requirement	Check	Design value	Limit	Status
H-05	HUL-D16	Box lid above the water at design mass	153 mm	≥ 150	PASS
H-06	HUL-D06, REC-001	Buoyancy with two segments breached, box flooded the two largest segments breached, 10% void each	7587 g	$\geq 1.5 \times 2100 = 3150$ g (and $\geq 1.5 \times 2465 = 3697$ g)	PASS
H-07	HUL-D08, MEC-004	Static heel, 300 g at the deck edge	2.1°	$\leq 10^\circ$	PASS
H-08	HUL-D09, MEC-004	Stability: max righting arm; angle of vanishing stability turn at 1.5 m/s, R 1 m heels $< 5^\circ$; confirmed in the pool (V-item)	GZ 110 mm at 15°; vanishes at 65°	vanishing $>$ wave slope 17° + edge heel 2°	PASS
H-09	REC-005, REC-D06	Tow by the hoop without capsize (sideways, worst case) a 50 N pull is a strength case; towed, the pull equals the drag	capsizes at 15 N side pull	$>$ side drag at 0.5 m/s ≈ 4.7 N	PASS

9.4 Structure

ID	Requirement	Check	Design value	Limit	Status
S-01	HUL-D21, MEC-014	Carry handle from the centre of gravity (x; y of the grip centre) each grip either side of the mast socket	39 mm; 48 mm	≤ 50	PASS
S-02	HUL-D21	Aft beam at 3 $\times$ design weight on the handle 73 N at the grip, hulls hanging from the beam ends	21 MPa	6063-T6 yield 160 MPa	PASS
S-03	IF-16, HUL-D14, HUL-D25	Segment joint: bending before it opens (hand-tight), vs 3 $\times$ design weight, bow and stern supported 2 $\times$ M12 at 97 N each from 20 N at the Ø35 knob ($K = 0.3$); load test B-item	15.5 N·m	≥ 3.4 N·m	PASS
S-04	IF-19, REC-D06	Moment at the mast socket, 50 N on the hoop the hoop is low on the mast: 5 $\times$ lower than the ICD's 15 N·m	4.3 N·m	design 30 N·m (IF-19)	PASS
S-05	REC-005, REC-D06	Hoop: internal size; 50 N pull the ring shares the load round both sides; pull test B-item	Ø64 mm; 29 MPa (cantilever, conservative)	$\geq \text{Ø}60$; PETG ≈ 50 MPa	PASS

9.5 Propulsion guarding and mount

ID	Requirement	Check	Design value	Limit	Status
P-01	MEC-010, PRP-D09	Largest opening a probe could enter (inlet annulus and rear guard) the duct is solid; the prop sits between inlet and guard; B-01 probe test	5.75 mm	< 8 mm	PASS
P-02	PRP-D10	Strut sweep; inlet vane angle	32°; 45°	$\geq 30^\circ$; $\geq 45^\circ$	PASS
P-03	IF-17, PRP-D12	Pod mount: dovetail angle, size, clearance; tool-free latch swap time is a bench test	60°, 20 $\times$ 40, 0.3 mm; tongue and detent	60°, 20 $\times$ 40, 0.3 mm	PASS
P-04	PRP-D14, CHD-003	Prop unreachable fitted or unfitted	guard is on the pod; screws need a tool	adult action only	PASS

9.6 Mast, GNSS and flag

ID	Requirement	Check	Design value	Limit	Status
E-01	MEC-013, HUL-D18, IF-19	GNSS to power wiring, box and motor leads motor leads run down from the glands, along the aft beam and the hull decks	252 mm	≥ 150	PASS
E-02	IF-19	Masthead top above the water (design mass) the ≤ 500 mm height from the pod bottom sets it	415 mm	≈ 450 , total height ≤ 500	FAIL CR-10
E-03	REC-D02, REC-003	Flag size; flag top above the water; below the GNSS	120 × 80; 326 mm	$\geq 120 \times 80$; ≥ 320 mm	PASS

9.7 Interfaces

ID	Requirement	Check	Design value	Limit	Status
I-01	IF-18, HUL-D15	Box inside size; glands; key dock the chosen box fixes these (TBC-13); submersion test B-item	197 × 127 × 83; 4 × PG7 at 25 mm on the aft face; cup on the lid	$\geq 180 \times 110 \times 70$	PASS
I-02	IF-20, HUL-D17, MEC-009	DUPLO grid stud size by test print against genuine bricks (TBC-14)	6 × 8 at 16 mm, Ø9.4 × 4.5	6 × 8 at 16 mm	PASS
I-03	HUL-D19	Camera hood overhang beyond the window	22 mm	≥ 15	PASS
I-04	HUL-D24	Box and battery bay open only by an adult force measured on the bench	2 over-centre latches, detent ≥ 40 N; box clips under them	≥ 40 N or two hands	TEST

9.8 Child safety

ID	Requirement	Check	Design value	Limit	Status
C-01	CHD-001, HUL-D22	Child-handled parts that fit the small-parts cylinder parts that do fit are bonded, captive or behind a tool: Joint dowel Ø8 × 20 (bonded); Beam end cap (bonded); Over-centre latch; Latch keeper (bonded to box); PG7 cable gland; Propeller Ø35, 3 blades; Camera OV5647 (reference)	none	none	PASS
C-02	CHD-002, HUL-D23	Edges and finger traps check on the first prints (inspection)	child-handled edges filleted or chamfered ≥ 0.6 mm; joints close face to face	≥ 0.6 mm	TEST

9.9 Visibility

ID	Requirement	Check	Design value	Limit	Status
V-01	REC-D01, REC-002	Fluorescent share of the top view segments printed in fluorescent orange; DUPLO plate green (bricks cover it)	64%	$\geq 50\%$	PASS

10. Proposed change requests

These are proposals. Nothing baselined changes until the owner accepts a CR at the CDR; then the documents named are reissued and the register is updated in the same commit.

CR-09 Mass limit and allocations

Finding	M-01: 2165 g against MEC-002's 1800 g. M-02: hull and structure 1551 g against HUL-D03's 1000 g. Segments, foam and joints 930 g against the ADD's 500 g.
Cause	The ADD mass table was an estimate made before any hull lines existed. The skin area of two 569 mm hulls with 50 mm of freeboard is about 0.4 m ² ; no printed shell that meets HUL-D10 weighs much less than 500 g, and the foam adds about 200 g.
What still passes	Every requirement that the mass affects has been checked at the real mass: draft $34.4 \leq 35$, freeboard $51 \geq 50$, heel $2.1^\circ \leq 10^\circ$, trim within $\pm 3^\circ$, reserve buoyancy $3 \times$. So what exceeding the limit costs is carrying weight and draft margin, not safety.
Proposal	Raise MEC-002 to 2.2 kg and the design mass to 2.5 kg (ICD load cases, IF-16 to IF-19). Set HUL-D03 to 1600 g and reissue the ADD mass table from this document. Keep draft ≤ 35 mm and freeboard ≥ 50 mm as they are; they are what the owner sees.
Alternatives	(a) EPS instead of XPS foam: about -90 g, more water uptake if breached. (b) Drop the tracker pocket and cap: -20 g. (c) Carbon mast and beams: about -80 g, +cost. (d) Shorter, deeper hulls: less skin, but draft rises past 35 mm. Even (a)+(b)+(c) leaves the boat near 2.0 kg, so the limit has to move either way.
Impact	SRS MEC-002; SSS-HUL budget and HUL-D03; SSS-PRP budget unchanged; ADD A.MASS; ICD design mass; sim model mass (KCL MASS_KG) and the SITL runs that use it.

CR-10 Masthead height (IF-19)

Finding	E-02: the masthead top is 415 mm above the water; IF-19 says ≈ 450 . The height limit (≤ 500 mm, MEC-001 and HUL-D01) is measured from the lowest point, which is the pod duct 47 mm below the keel. Overall height is 492 mm.
Proposal	Change IF-19 to 'GNSS top ≥ 400 mm above the waterline'. Nothing depends on the exact height: MEC-013's separation is 252 mm, and the flag, below the GNSS, still flies above 320 mm.
Impact	ICD IF-19 text only.

CR-11 Hull shell thickness (HUL-D10)

Finding	HUL-D10 asks for ≥ 3 perimeters and a shell ≥ 1.6 mm. With a 0.4 mm nozzle, 3 perimeters is 1.2 mm; 1.6 mm needs 4. The design uses 1.2 mm ASA.
Rationale	The shell is the skin of a foam sandwich, not a free panel: the foam carries it against impact and water pressure, and a breach does not sink the segment. 1.6 mm adds about 150 g, which CR-09 cannot afford.
Proposal	HUL-D10: 'shell ≥ 3 perimeters (≥ 1.2 mm) over a bonded foam core'. Verify with the IF-16 joint load test and a drop test of one mid segment onto gravel from 0.5 m.
Impact	SSS-HUL HUL-D10; FMEA hull-breach rows (the occurrence estimate, not the controls).

10.1 TBCs this design closes

TBC	Was	Now
TBC-11 (IF-16)	Section 90 × 100 at the joint	84 × 90, lines in the CAD
TBC-13 (IF-18)	Box and saddle	Any clip-lock box within 200 × 130 × 86 external; the saddle is drawn for it. Buy the box, then set BOX_* and rebuild.
TBC-14 (IF-20)	Stud size	Still open: needs test prints against genuine DUPLO
TBC-12 (IF-17)	Pod connector	Still open: the CAD reserves the parking clips and the lead route only

11. Manufacture, assembly and verification

11.1 Printing

Parts	Material	Settings	Orientation
Hull segments	Fluorescent orange ASA	0.4 nozzle, 3 perimeters, 4 top/bottom, no infill needed (shell), enclosure	Open joint end down (mid: either end); tree supports inside the stern's transom, removed through the open end before foaming
Thumb-screws, dowels	PETG	4 perimeters, 40% infill, 0.15 mm layers for the thread	Axis vertical, knob down
Pod body, guard, prop	PETG (black)	3 perimeters, 40% infill	Pod: back face down, supports under the duct; prop: hub down
Tray, handle, hood, collar, plate	PETG (fluorescent where hi-vis)	3 perimeters, 25% gyroid	Largest flat face down
Masthead	Translucent PETG	2 perimeters in the diffuser ring	Dome up

11.2 Stock and bought items

Item	Qty	Note
6063 Al rectangular tube 20 × 10 × 1.5	0.6 m	two 300 mm beams, drilled to the rail grid
Al round tube Ø16 × 1	0.33 m	mast; one Ø4.2 hole for the collar bolt
XPS board 50 mm, closed-cell	≈ 7 L	profiled cores
Clip-lock box, 1.2-1.5 L	1	TBC-13
PG7 glands	4	IF-18
Acrylic 2 mm disc Ø40	1	camera window
Ripstop, fluorescent	120 × 80	flag
M4 heat-set inserts; M4 × 16 A4	8; 8	beams
M3 × 16/20 A4, nylon-insert nuts	8	tray and handle through the rail
M3 × 30 A4 pins	2	latch pivots
M4 × 30 A4 and nut	1	collar cross-bolt
M2.5 × 8 A2; M2 × 6 A2	4; 8	guards; motor mounts
M3 inserts and M3 × 8 A2; Ø1 O-ring cord	6; 6	tracker pocket caps

The stock items are new to the bill of materials; they are to be priced into the Key Component List before the CDR and checked against the cap (open item O-3).

11.3 Assembly

Adult, once: foam the segments; bond dowels, caps, keepers and the key dock; fit inserts; bolt the beams to the mid segments and the tray and handle to the beams; fit glands, window and hood to the box; fit the motors and props, then the guards. **Crew, each session** (no tools, ≤ 20 N): join the bow and stern segments to the core with the thumb-screws; slide on the pods until they click; plug in the pod leads (adult checks); put the mast in its socket and tighten its knob; plug in the flag; clip on the DUPLO plate.

11.4 Verification of the first prints

ID	What	Pass
M-V1	Weigh each segment before and after foaming	foam fill ≥ 90% of the cavity (HUL-D05)
M-V2	Weigh the boat ready to sail	matches section 5 within 5%
M-V3	Joint load test: two joined segments, 3 × design weight at mid-span (IF-16)	no opening or cracking

ID	What	Pass
M-V4	Probe test on the inlet and guard (MEC-010)	8 mm probe cannot touch a blade
M-V5	Hoop pull 50 N; tow in the pool (REC-005)	no damage; no capsize
M-V6	Latch release force (HUL-D24)	≥ 40 N
M-V7	Pod swap timed (MEC-008), pull test (IF-17)	< 2 min; holds 30 N
M-V8	Stud fit with genuine bricks (IF-20)	2×2 stays on at 45° and with shake; child removes it
M-V9	Box submersion 30 min at 0.3 m (MEC-012)	tissue dry
M-V10	Float test: draft, trim, heel with 300 g at the edge	within 3 mm and 1° of section 6
M-V11	Edges and small parts inspection (CHD-001/002)	no sharp edges; no loose small parts

11.5 Open items

ID	Item
O-1	Owner decisions on CR-09, CR-10, CR-11 at the CDR
O-2	Choose the box; rebuild with its dimensions (TBC-13)
O-3	Price the stock items into the KCL and check the cap
O-4	Pod connector part number and parking clip (TBC-12)
O-5	Update the simulator's mass and inertia from section 5 when CR-09 is decided
O-6	Motor lead route through the strut: groove and potting detail after the connector is chosen

Appendix A. Part list and files

The zip holds these files, the assembly (STEP with colours, and GLB), the CadQuery source and the results file. Part files are moved to the origin; the assembly places them.

Part	Name	Qty	Files (in the CAD zip)
HUL-101	Bow segment	2	parts/HUL-101_bow_segment.step parts/HUL-101_bow_segment.stl
HUL-102	Mid segment	2	parts/HUL-102_mid_segment.step parts/HUL-102_mid_segment.stl
HUL-103	Stern segment	2	parts/HUL-103_stern_segment.step parts/HUL-103_stern_segment.stl
HUL-104	Joint thumb-screw M12	8	parts/HUL-104_joint_thumb_screw_m12.step parts/HUL-104_joint_thumb_screw_m12.stl
HUL-105	Joint dowel Ø8 × 20 (bonded)	8	parts/HUL-105_joint_dowel_8_20_bonded.step parts/HUL-105_joint_dowel_8_20_bonded.stl
HUL-106	Tracker pocket cap	2	parts/HUL-106_tracker_pocket_cap.step parts/HUL-106_tracker_pocket_cap.stl
HUL-201	Crossbeam 20 × 10 × 1.5 Al tube, 300 long	2	parts/HUL-201_crossbeam_20_10_1_5_al_tube_300_long.step
HUL-202	Beam end cap (bonded)	4	parts/HUL-202_beam_end_cap_bonded.step parts/HUL-202_beam_end_cap_bonded.stl
HUL-301	Electronics box body (clip-lock, modified)	1	parts/HUL-301_electronics_box_body_clip_lock_modified.step
HUL-302	Electronics box lid	1	parts/HUL-302_electronics_box_lid.step
HUL-303	Saddle tray	1	parts/HUL-303_saddle_tray.step parts/HUL-303_saddle_tray.stl
HUL-304	Over-centre latch	2	parts/HUL-304_over_centre_latch.step parts/HUL-304_over_centre_latch.stl
HUL-305	Latch keeper (bonded to box)	2	parts/HUL-305_latch_keeper_bonded_to_box.step parts/HUL-305_latch_keeper_bonded_to_box.stl
HUL-306	PG7 cable gland	4	parts/HUL-306_pg7_cable_gland.step
HUL-307	Key dock (bonded to lid)	1	parts/HUL-307_key_dock_bonded_to_lid.step parts/HUL-307_key_dock_bonded_to_lid.stl
HUL-308	Arming key fob (magnet bonded in)	1	parts/HUL-308_arming_key_fob_magnet_bonded_in.step parts/HUL-308_arming_key_fob_magnet_bonded_in.stl
HUL-401	DUPLO deck plate	1	parts/HUL-401_duplo_deck_plate.step parts/HUL-401_duplo_deck_plate.stl
HUL-402	Mast step and carry handle	1	parts/HUL-402_mast_step_and_carry_handle.step parts/HUL-402_mast_step_and_carry_handle.stl
HUL-403	Mast tube Ø16 × 1 Al, 311 long	1	parts/HUL-403_mast_tube_16_1_al_311_long.step
HUL-404	Mast thumb-screw M8 (captive)	1	parts/HUL-404_mast_thumb_screw_m8_captive.step parts/HUL-404_mast_thumb_screw_m8_captive.stl
HUL-501	Camera hood	1	parts/HUL-501_camera_hood.step parts/HUL-501_camera_hood.stl
HUL-502	Camera window, 2 mm acrylic	1	parts/HUL-502_camera_window_2_mm_acrylic.step
REC-101	Collar and recovery hoop	1	parts/REC-101_collar_and_recovery_hoop.step parts/REC-101_collar_and_recovery_hoop.stl
REC-102	Flag staff with ball cap	1	parts/REC-102_flag_staff_with_ball_cap.step parts/REC-102_flag_staff_with_ball_cap.stl
REC-103	Flag, fluorescent ripstop 120 × 80	1	parts/REC-103_flag_fluorescent_ripstop_120_80.step

Part	Name	Qty	Files (in the CAD zip)
REC-104	Masthead: GNSS dome and LED diffuser	1	parts/REC-104_masthead_gnss_dome_and_led_diffuser.step parts/REC-104_masthead_gnss_dome_and_led_diffuser.stl
PRP-101	Thruster pod body	2	parts/PRP-101_thruster_pod_body.step parts/PRP-101_thruster_pod_body.stl
PRP-102	Rear guard	2	parts/PRP-102_rear_guard.step parts/PRP-102_rear_guard.stl
PRP-103	Propeller Ø35, 3 blades	2	parts/PRP-103_propeller_35_3_blades.step parts/PRP-103_propeller_35_3_blades.stl
KC-04	Motor 2205 (reference)	2	parts/KC-04_motor_2205_reference.step
KC-11	Camera OV5647 (reference)	1	parts/KC-11_camera_ov5647_reference.step